Construction Chemicals

SUBMITTAL DOCUMENT

Bondmix EP

Solvent-Free Epoxy
Bonding Agent — Old to New Concrete



Bond	Tensile	Solvent-Free	Pot Life
≥ 2.0 MPa	≥ 15 MPa	100 % solids	30 – 45 min

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Bondmix EP — Solvent-Free Epoxy Bonding Agent — Old to New Concrete

Project Name		**Document No.**	OMX-MS-BONDMIXEP
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Bondmix EP	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Bondmix EP on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers surface preparation, mixing and application of Bondmix EP epoxy bonding agent for bonding fresh concrete or mortar to hardened concrete.

- Inspection and acceptance of the substrate and the work area.
- Surface preparation and repair of defects.
- Priming of the prepared surface.
- Mixing and application of the material as described in this document.
- Curing, protection and handover of the completed area.
- Quality control, inspection and documentation of each stage.

3. Definitions

Term	Meaning
Pot life	Time the mixed material remains workable after Part A and Part B are combined.
Recoat interval	Minimum and maximum time between coats to achieve inter-coat adhesion.
WFT / DFT	Wet film thickness measured during application / dry film thickness after cure.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Kit / set	Pre-weighed Part A and Part B supplied to be mixed completely.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.

Role	Responsibility
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Bondmix EP	Epoxy bonding agent, 5 and 20 kg sets

5.1 Key product data

Property	Typical Value
Mixing ratio (A : B)	2 : 1 by weight
Density	1.40 – 1.50 kg/L
Pot life	30 – 45 min
Tensile strength	≥ 15 MPa
Bond strength	≥ 2.0 MPa
Application temperature	+10 °C to +35 °C

6. Tools & Equipment

- Grinder / scabbler
- Wire brush
- Vacuum
- Slow-speed mixer
- Stiff brushes
- Rollers
- Spatulas
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Wet-film thickness gauge
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Goggles
- Overalls

7. Storage & Handling

- 12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep Part A and Part B of each kit together and do not mix parts from different batches.
- Store away from heat, sparks and ignition sources.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Remove laitance and weak concrete by scabbling or grinding.
2. Clean the surface of dust and oil; dry or matt-damp.
3. Prepare formwork and fresh concrete so it can be placed within the open time.
4. Measure substrate temperature, air temperature and relative humidity; confirm the substrate is at least 3 °C above dew point.
5. Protect adjacent surfaces, drains and fixtures with masking tape and sheeting.
6. Obtain QA/QC approval of the prepared surface before starting the next step.

10. Mixing

1. Condition materials to between +15 °C and +25 °C before mixing.
2. Check the batch numbers and expiry dates of both parts and record them.
3. Add Part B to Part A (2 : 1 by weight).
4. Mix 2 – 3 minutes with a slow-speed mixer until uniform.
5. Mix only the quantity that can be applied within the pot life.

11. Application

1. Brush or roll the bonding agent well into the surface at 0.30 – 0.60 kg/m².
2. Place the fresh concrete or mortar while the coat is still tacky.
3. Compact the fresh concrete well against the bonding layer.
4. Work in manageable bays, keeping a wet edge to avoid lap marks.
5. Check consumption by area and wet film thickness regularly during application and record the results.

12. Curing & Protection

- Pot life 30 – 45 minutes.
- Cure the fresh concrete as normal.
- Protect the applied area from water, dust, other trades and traffic until the stated cure times are reached.
- Barricade the area and display warning signs.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Substrate	Before application	Sound, clean, laitance removed	QA/QC
Mixing	Each set	2 : 1, uniform	Site Engineer
Consumption	During application	0.30 – 0.60 kg/m²	QA/QC
Placement	During concreting	Fresh concrete placed on tacky coat	Site Engineer

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report
Climate	Air / substrate temperature, RH, dew point	Start of shift and every 4 hours	Daily climate log
Surface preparation	Visual, moisture, soundness	Each area	Inspection request (hold point)
Application	Consumption, thickness, appearance	Each coat / area	Daily progress report
Final inspection	Visual, continuity, finish	Each completed area	Handover checklist

14. Defects & Remedies

Defect	Likely Cause	Remedy
Pinholes / bubbles	Outgassing from porous substrate; air entrapped in mixing	Seal with primer; mix at low speed; de-aerate with spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash and abrade; recoat in suitable conditions

Defect	Likely Cause	Remedy
Poor adhesion	Contaminated, weak or damp substrate; recoat window exceeded	Remove, re-prepare surface and reapply
Soft spots	Incorrect mixing or ratio	Remove affected area and reapply with full kits
Uneven colour / lap marks	Loss of wet edge; different batches	Abrade and apply a further coat from one batch

15. Health, Safety & Environment

Hazard	Control Measures
Skin and eye contact	Gloves, goggles, overalls; eye-wash available
Inhalation of vapours / dust	Ventilation; masks as required
Slips and trips	Housekeeping; barricades and signs
Manual handling	Correct lifting technique; mechanical aids
Electrical tools	Inspected tools; protected cables

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.
- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with epoxy thinner before the material cures.

17. Limitations

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate clean, sound, dry or matt-damp.
- Do not apply in rain or when rain is expected before the material has set.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Surface prepared	☐ Yes ☐ No	
Complete sets mixed	☐ Yes ☐ No	
Bonding agent applied	☐ Yes ☐ No	
Concrete placed while tacky	☐ Yes ☐ No	
Concrete compacted and cured	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Bondmix EP — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Concrete Admixture

Bondmix EP

Solvent-Free Epoxy
Bonding Agent — Old to New Concrete



Product Description

Bondmix EP is a two-component, solvent-free epoxy bonding agent for bonding fresh concrete or mortar to hardened concrete. It is applied by brush, roller or spatula and the new concrete is placed while the bonding agent is still tacky.

Uses

- Bonding fresh to hardened concrete.
- Structural repairs and patching.
- Industrial floor toppings.
- Construction joints in concrete.

Advantages

- Bond strength ≥ 2.0 MPa.
- Tensile strength ≥ 15 MPa.
- Solvent-free.
- Pot life 30 – 45 min.
- Brush, roller or spatula application.

Application

1. **Surface Preparation**
   Remove laitance and weak concrete; clean, dry or matt-damp surface.
2. **Mixing**
   Add Part B to Part A (2 : 1 by weight) and mix 2 – 3 min.
3. **Application**
   Apply 0.30 – 0.60 kg/m² and place fresh concrete while still tacky.

Consumption

0.30 – 0.60 kg/m²
depending on roughness
Rougher surfaces need the higher rate.

Packaging

5 kg sets (3.33 kg A + 1.67 kg B) and 20 kg sets (13.33 kg A + 6.67 kg B).

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	2 : 1 by weight	—
Density	1.40 – 1.50 kg/L	—
Pot life	30 – 45 min	—
Tensile strength	≥ 15 MPa	—
Bond strength	≥ 2.0 MPa	—
Application temperature	+10 °C to +35 °C	—

Consumption

Surface	Consumption
Smooth	0.30 – 0.40 kg/m²
Rough	0.40 – 0.60 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate clean, sound, dry or matt-damp.
- Fresh concrete must be placed while the bonding agent is tacky.

Application Notes

- Mix complete sets.
- Work the bonding agent well into the surface.
- Place fresh concrete within the open (tacky) time.
- If the coat has cured, roughen and reapply.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with epoxy thinner before the material cures.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.